>> Importing Required Libraries

In [ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import accuracy_score, classification_report, confusion_m
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from xgboost import XGBClassifier
import warnings
warnings.filterwarnings('ignore')
```

>> Loading & Inspecting the Dataset

In [ ]:
```
df = pd.read_excel("/content/Obesity_Level_Estimation_Dataset.xlsx")
df.head()
```

Out[ ]:

	Age	Gender	Height	Weight	CALC	FAVC	FCVC	NCP	SCC	SMOKE	CH
0	21.0	Female	1.62	64.0	no	no	2.0	3.0	no	no	
1	21.0	Female	1.52	56.0	Sometimes	no	3.0	3.0	yes	yes	
2	23.0	Male	1.80	77.0	Frequently	no	2.0	3.0	no	no	
3	27.0	Male	1.80	87.0	Frequently	no	3.0	3.0	no	no	
4	22.0	Male	1.78	89.8	Sometimes	no	2.0	1.0	no	no	

In [ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2111 entries, 0 to 2110
Data columns (total 17 columns):
 #   Column                           Non-Null Count  Dtype
---  ------                           --------------  -----
 0   Age                              2111 non-null   float64
 1   Gender                           2111 non-null   object
 2   Height                           2111 non-null   float64
 3   Weight                           2111 non-null   float64
 4   CALC                             2111 non-null   object
 5   FAVC                             2111 non-null   object
 6   FCVC                             2111 non-null   float64
 7   NCP                              2111 non-null   float64
 8   SCC                              2111 non-null   object
 9   SMOKE                            2111 non-null   object
 10  CH2O                             2111 non-null   float64
 11  family_history_with_overweight   2111 non-null   object
 12  FAF                              2111 non-null   float64
 13  TUE                              2111 non-null   float64
 14  CAEC                             2111 non-null   object
 15  MTRANS                           2111 non-null   object
 16  NObeyesdad                       2111 non-null   object
dtypes: float64(8), object(9)
memory usage: 280.5+ KB
```

In [ ]:
```
df.isnull().sum()
```

Out[ ]:

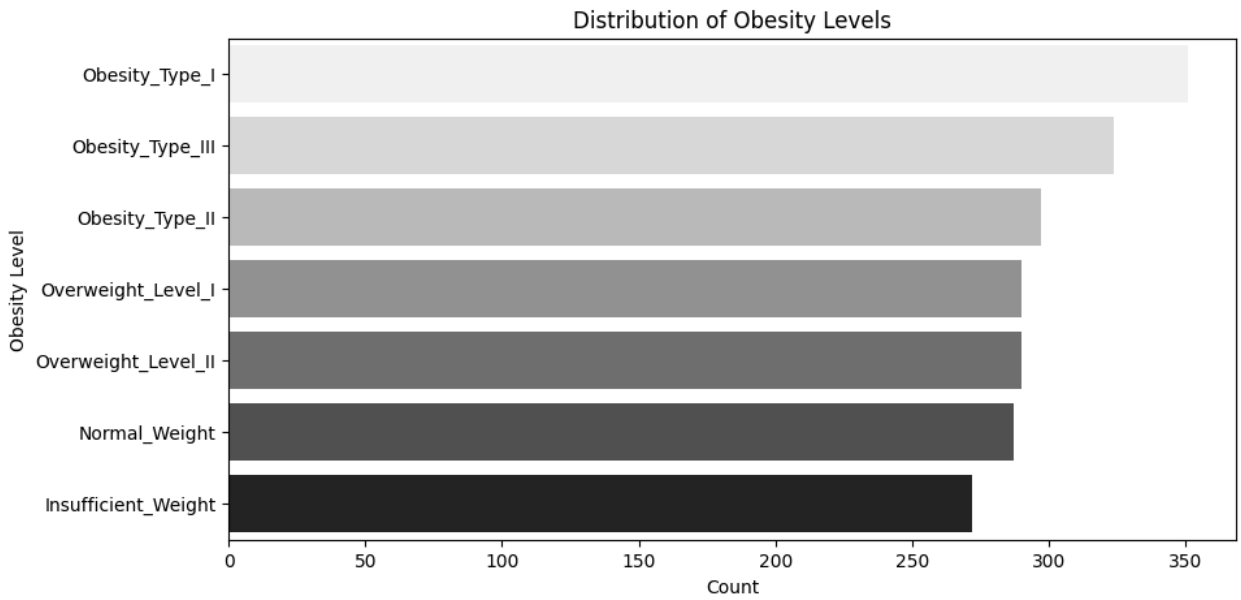
	0
Age	0
Gender	0
Height	0
Weight	0
CALC	0
FAVC	0
FCVC	0
NCP	0
SCC	0
SMOKE	0
CH2O	0
family_history_with_overweight	0
FAF	0
TUE	0
CAEC	0
MTRANS	0
NObeyesdad	0

dtype: int64

>> Exploratory Data Analysis (EDA)

In [ ]:

```
plt.figure(figsize=(10,5))
sns.countplot(y='NObeyesdad', data=df, order=df['NObeyesdad'].value_counts().i
plt.title("Distribution of Obesity Levels")
plt.xlabel("Count")
plt.ylabel("Obesity Level")
plt.show()
```



In [ ]:
```
df['NObeyesdad'].value_counts()
```

Out[ ]:

NObeyesdad	count
Obesity_Type_I	351
Obesity_Type_III	324
Obesity_Type_II	297
Overweight_Level_I	290
Overweight_Level_II	290
Normal_Weight	287
Insufficient_Weight	272

dtype: int64

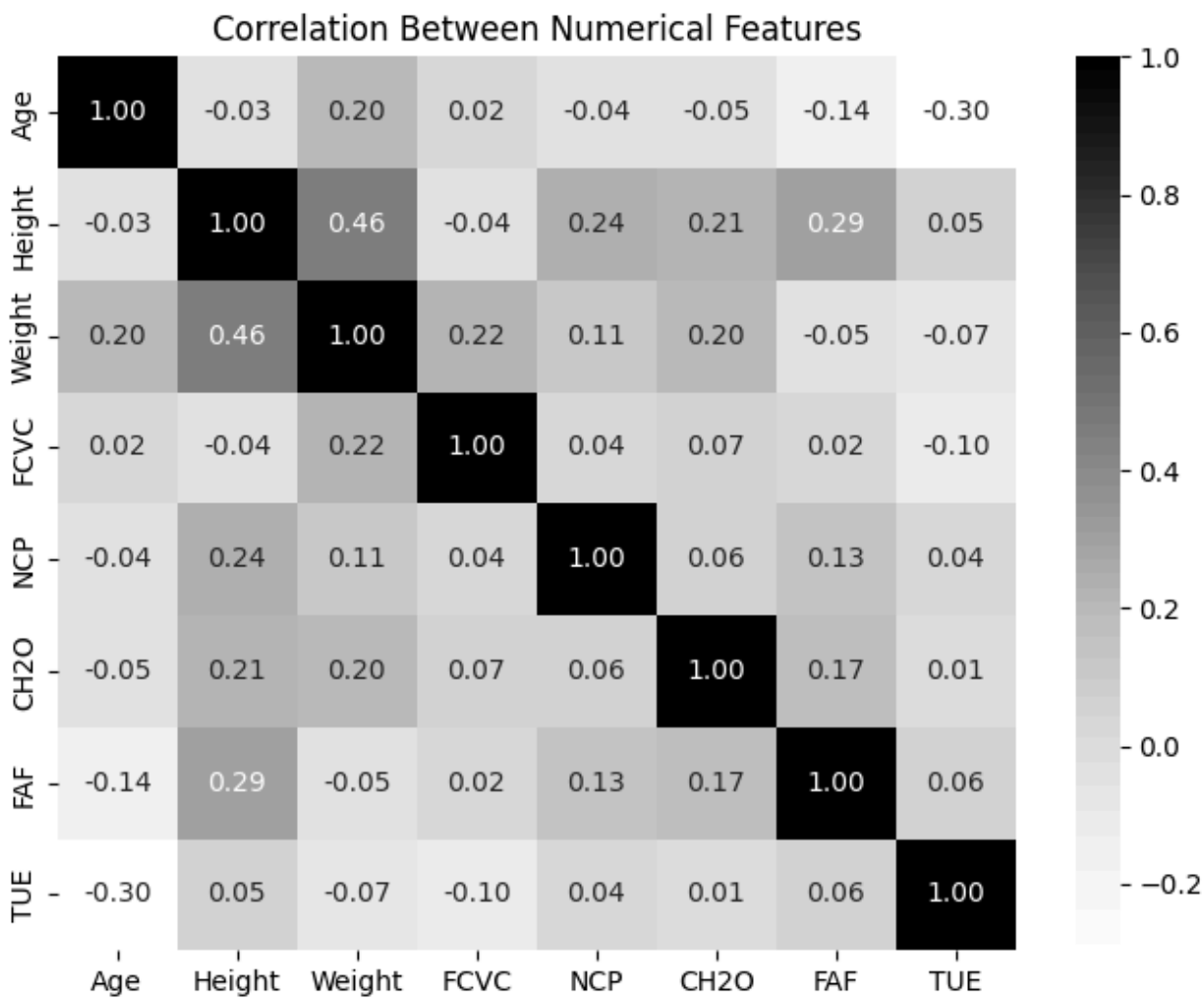
In [ ]:
```
df.describe()
```

Out[ ]:

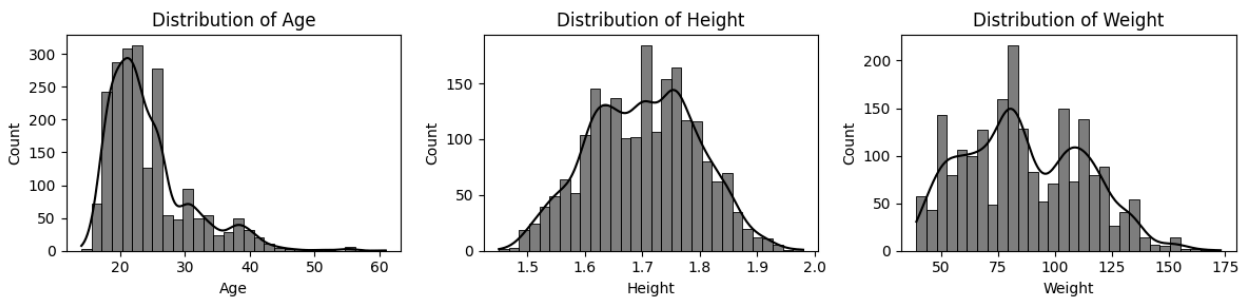
	Age	Height	Weight	FCVC	NCP	C
count	2111.000000	2111.000000	2111.000000	2111.000000	2111.000000	2111.00
mean	24.312600	1.701677	86.586058	2.419043	2.685628	2.00
std	6.345968	0.093305	26.191172	0.533927	0.778039	0.61
min	14.000000	1.450000	39.000000	1.000000	1.000000	1.00
25%	19.947192	1.630000	65.473343	2.000000	2.658738	1.58
50%	22.777890	1.700499	83.000000	2.385502	3.000000	2.00
75%	26.000000	1.768464	107.430682	3.000000	3.000000	2.47
max	61.000000	1.980000	173.000000	3.000000	4.000000	3.00

In [ ]:

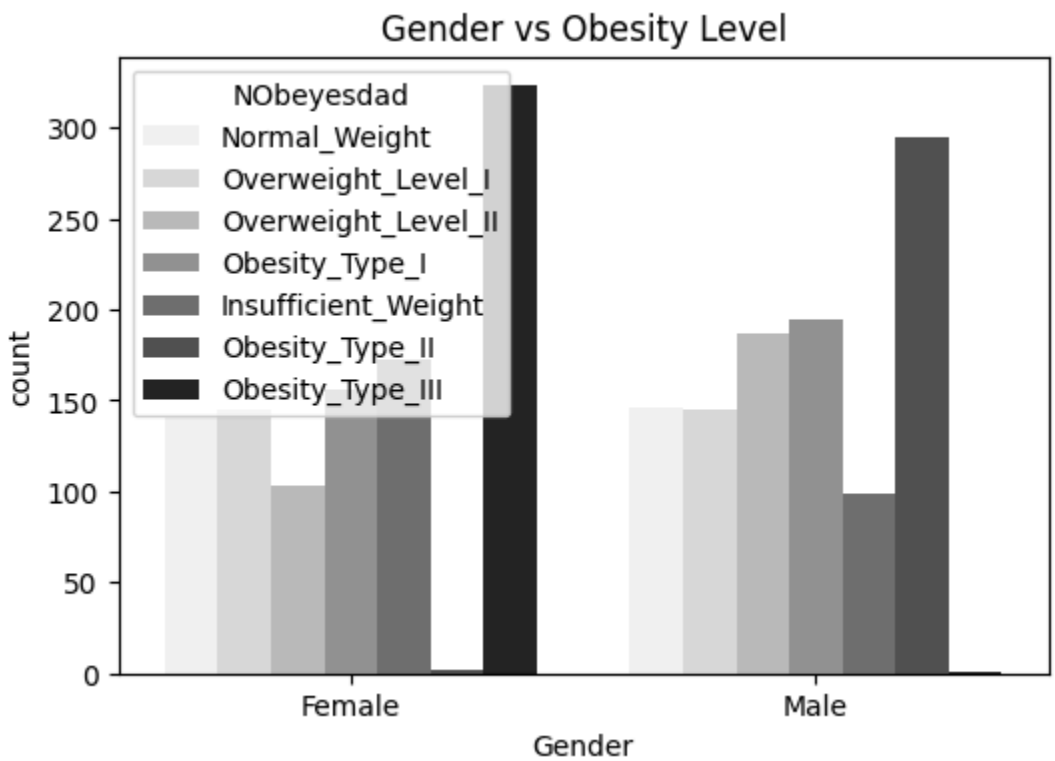
```
plt.figure(figsize=(8,6))
sns.heatmap(df.select_dtypes(include='number').corr(), annot=True, cmap='Greys
plt.title("Correlation Between Numerical Features")
plt.show()
```



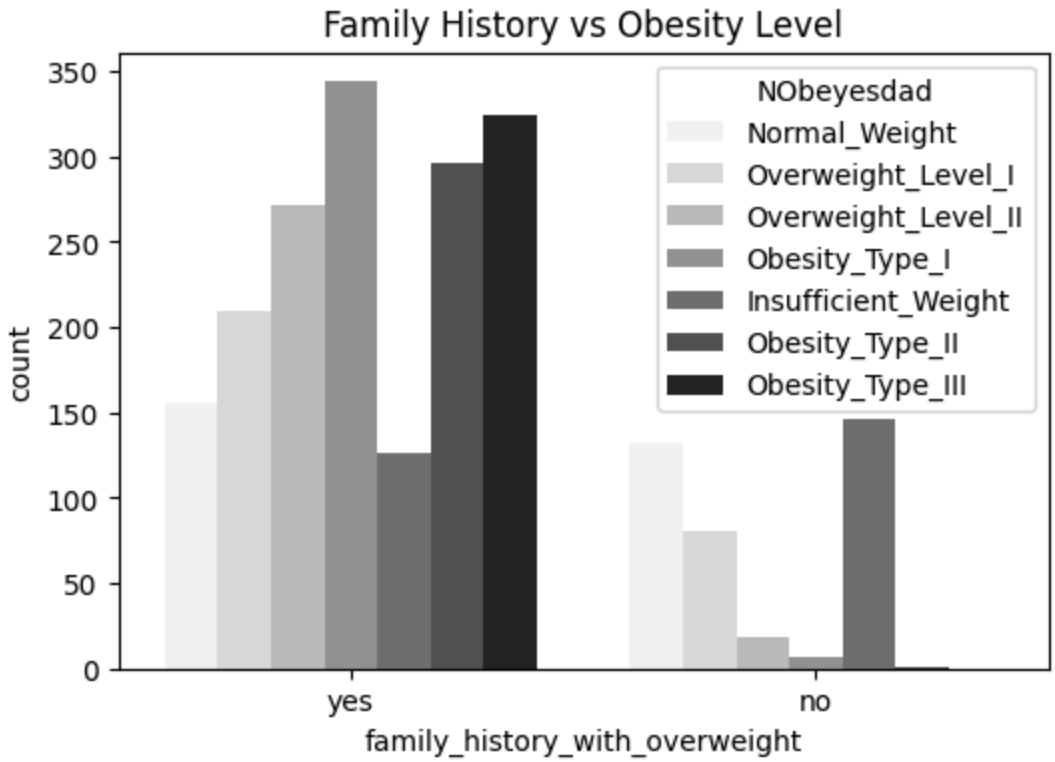
In [ ]:
```
fig, axes = plt.subplots(1, 3, figsize=(12, 3))
for i, col in enumerate(['Age', 'Height', 'Weight']):
    sns.histplot(df[col], kde=True, bins=30, color='black', ax=axes[i]).set_ti
plt.tight_layout()
plt.show()
```



In [ ]:
```
plt.figure(figsize=(6,4))
sns.countplot(x='Gender', hue='NObeyesdad', data=df, palette='Greys')
plt.title("Gender vs Obesity Level")
plt.show()
```



In [ ]:
```
plt.figure(figsize=(6,4))
sns.countplot(x='family_history_with_overweight', hue='NObeyesdad', data=df, p
plt.title("Family History vs Obesity Level")
plt.show()
```



>> Data Preprocessing

In [ ]:
```
cat_cols = df.select_dtypes(include='object').columns.tolist()
num_cols = df.select_dtypes(include='number').columns.tolist()
print("Categorical Columns:", cat_cols)
print("Numerical Columns:", num_cols)
```

```
Categorical Columns: ['Gender', 'CALC', 'FAVC', 'SCC', 'SMOKE', 'family_histor
y_with_overweight', 'CAEC', 'MTRANS', 'NObeyesdad']
Numerical Columns: ['Age', 'Height', 'Weight', 'FCVC', 'NCP', 'CH2O', 'FAF', 'T
UE']
```

In [ ]:
```
label_encoders = {}
for col in cat_cols:
    if col != 'NObeyesdad':
        le = LabelEncoder()
        df[col] = le.fit_transform(df[col])
        label_encoders[col] = le
```

In [ ]:
```
target_encoder = LabelEncoder()
df['NObeyesdad'] = target_encoder.fit_transform(df['NObeyesdad'])
```

In [ ]:
```
X = df.drop('NObeyesdad', axis=1)
y = df['NObeyesdad']
```

In [ ]:
```
scaler = StandardScaler()
X[num_cols] = scaler.fit_transform(X[num_cols])
```

In [ ]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

In [ ]:
```
print(f"Training samples: {X_train.shape[0]}, Testing samples: {X_test.shape[0
print(f"Feature count: {X_train.shape[1]}")
```

```
Training samples: 1688, Testing samples: 423
Feature count: 16
```

OBJECTIVE 1 > MULTI-CLASS OBESITY LEVEL CLASSIFICATION

Algorithm 1 >> Support Vector Machine (SVM)

In [ ]:
```
svm_model = SVC(kernel='rbf', C=10, gamma='scale', probability=True, random_st
svm_model.fit(X_train, y_train)
```

Out[ ]:
```
▾ SVC
SVC(C=10, probability=True, random_state=42)
```

In [ ]:
```
y_pred_svm = svm_model.predict(X_test)
```

In [ ]:
```
acc_svm = accuracy_score(y_test, y_pred_svm)
print(f"SVM Accuracy: {acc_svm:.4f}")
```

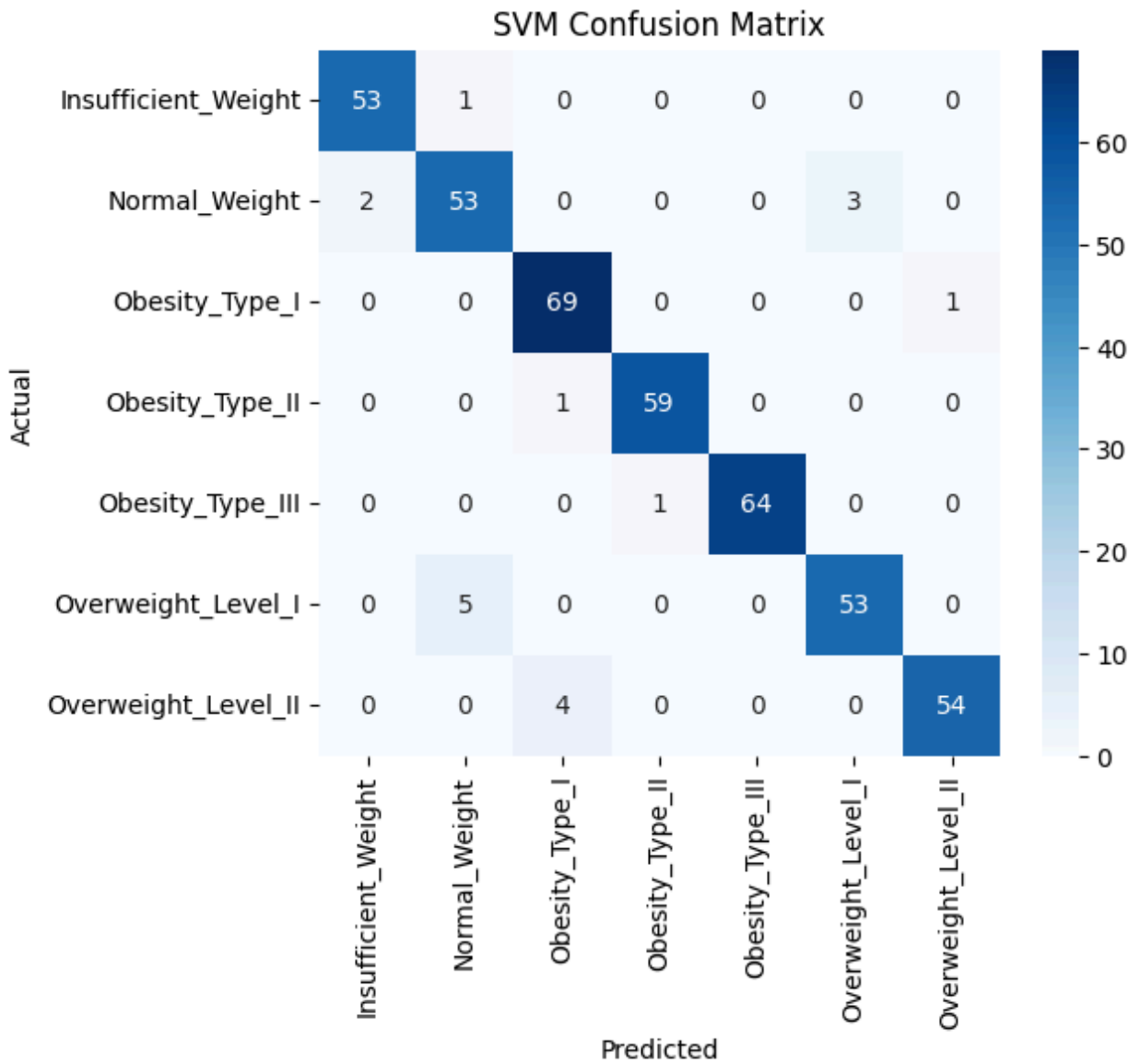
```
SVM Accuracy: 0.9574
```

In [ ]:
```
print("Classification Report:")
print(classification_report(y_test, y_pred_svm, target_names=target_encoder.cl
```

```
Classification Report:
                     precision    recall  f1-score   support

Insufficient_Weight       0.96      0.98      0.97        54
      Normal_Weight       0.90      0.91      0.91        58
     Obesity_Type_I       0.93      0.99      0.96        70
    Obesity_Type_II       0.98      0.98      0.98        60
   Obesity_Type_III       1.00      0.98      0.99        65
 Overweight_Level_I       0.95      0.91      0.93        58
Overweight_Level_II       0.98      0.93      0.96        58

           accuracy                           0.96       423
          macro avg       0.96      0.96      0.96       423
       weighted avg       0.96      0.96      0.96       423
```

In [ ]:
```
cm_svm = confusion_matrix(y_test, y_pred_svm)
```

```
plt.figure(figsize=(6,5))
sns.heatmap(cm_svm, annot=True, fmt='d', cmap='Blues',
            xticklabels=target_encoder.classes_,
            yticklabels=target_encoder.classes_)
plt.title("SVM Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



Algorithm 2 >> Random Forest (RF)

In [ ]:
```
rf_model = RandomForestClassifier(
    n_estimators=300,
    max_depth=None,
    min_samples_split=2,
    min_samples_leaf=1,
    random_state=42
```

```
)
rf_model.fit(X_train, y_train)
```

Out[ ]:

```
▾ RandomForestClassifier
RandomForestClassifier(n_estimators=300, random_state=42)
```

In [ ]:

```
y_pred_rf = rf_model.predict(X_test)
```

In [ ]:

```
acc_rf = accuracy_score(y_test, y_pred_rf)
print(f"Random Forest Accuracy: {acc_rf:.4f}")
```

```
Random Forest Accuracy: 0.9574
```

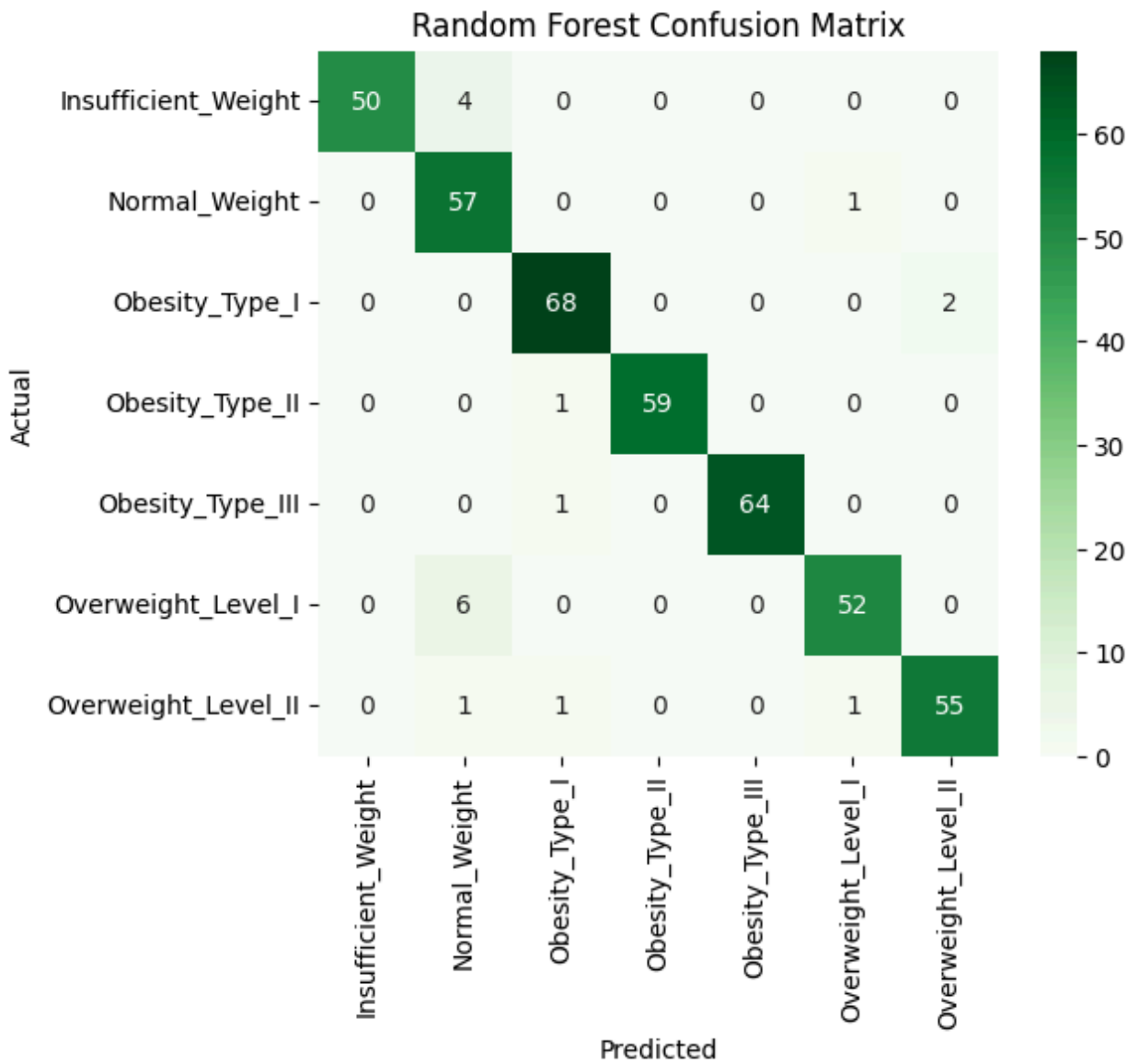
In [ ]:

```
print("Classification Report:")
print(classification_report(y_test, y_pred_rf, target_names=target_encoder.cla
```

```
Classification Report:
                     precision    recall  f1-score   support

Insufficient_Weight       1.00      0.93      0.96        54
      Normal_Weight       0.84      0.98      0.90        58
     Obesity_Type_I       0.96      0.97      0.96        70
    Obesity_Type_II       1.00      0.98      0.99        60
   Obesity_Type_III       1.00      0.98      0.99        65
 Overweight_Level_I       0.96      0.90      0.93        58
Overweight_Level_II       0.96      0.95      0.96        58

           accuracy                           0.96       423
          macro avg       0.96      0.96      0.96       423
       weighted avg       0.96      0.96      0.96       423
```

In [ ]:

```
cm_rf = confusion_matrix(y_test, y_pred_rf)
plt.figure(figsize=(6,5))
sns.heatmap(cm_rf, annot=True, fmt='d', cmap='Greens',
            xticklabels=target_encoder.classes_,
            yticklabels=target_encoder.classes_)
plt.title("Random Forest Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



Algorithm 3 >> XGBoost

In [ ]:
```
xgb_model = XGBClassifier(
    n_estimators=400,
    learning_rate=0.05,
    max_depth=6,
    subsample=0.9,
    colsample_bytree=0.9,
    objective='multi:softmax',
    num_class=len(target_encoder.classes_),
    eval_metric='mlogloss',
    random_state=42
)
```

In [ ]:
```
xgb_model.fit(X_train, y_train)
```

Out[ ]:

```
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=0.9, device=None, early_stopping_round
s=None,
              enable_categorical=False, eval_metric='mlogloss',
              feature_types=None, feature_weights=None, gamma=None,
              grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=0.05, max_b
in=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
```

In [ ]:

```
y_pred_xgb = xgb_model.predict(X_test)
```

In [ ]:

```
acc_xgb = accuracy_score(y_test, y_pred_xgb)
print(f"XGBoost Accuracy: {acc_xgb:.4f}")
```

```
XGBoost Accuracy: 0.9622
```

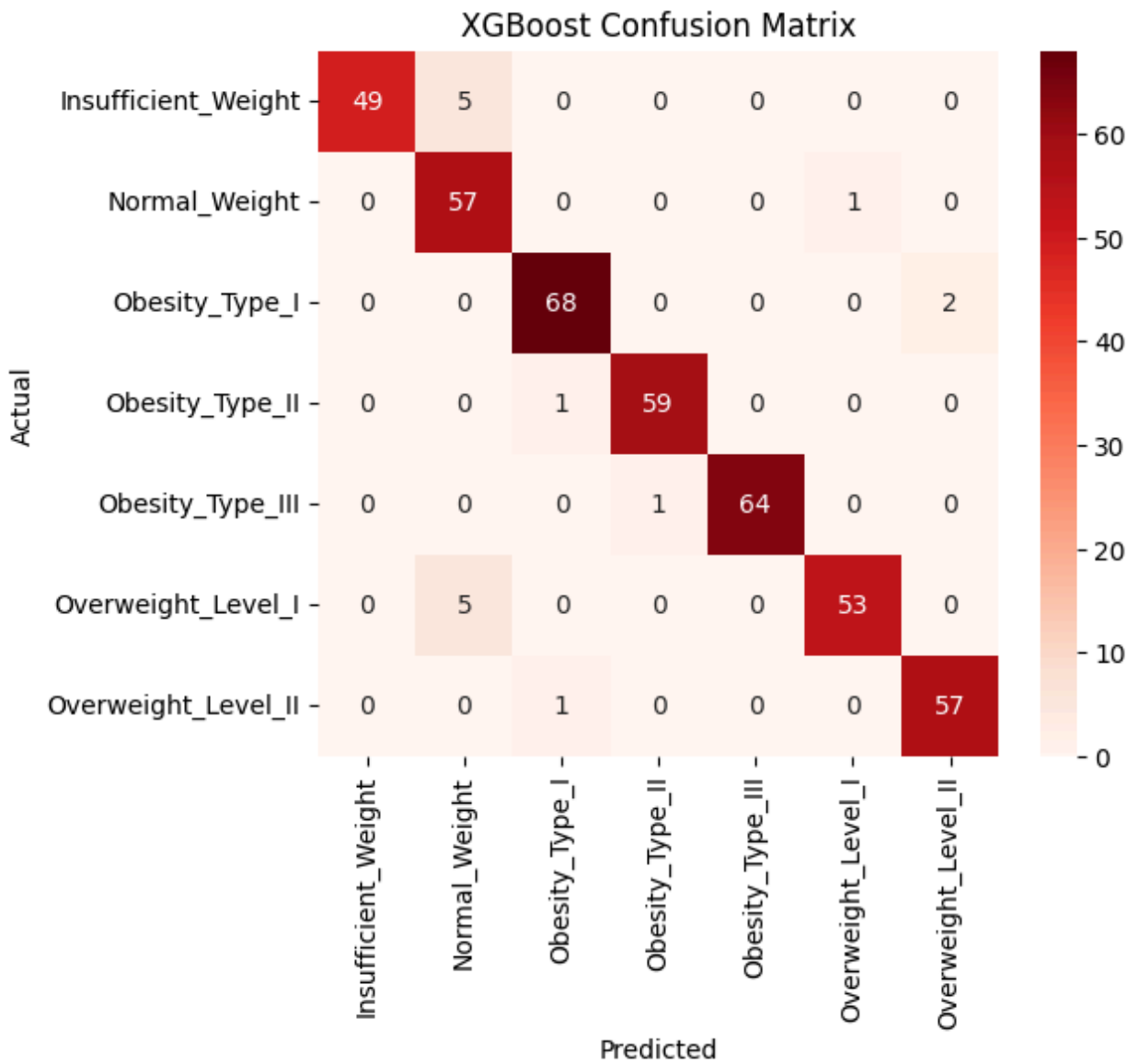
In [ ]:

```
print("Classification Report:")
print(classification_report(y_test, y_pred_xgb, target_names=target_encoder.cl
```

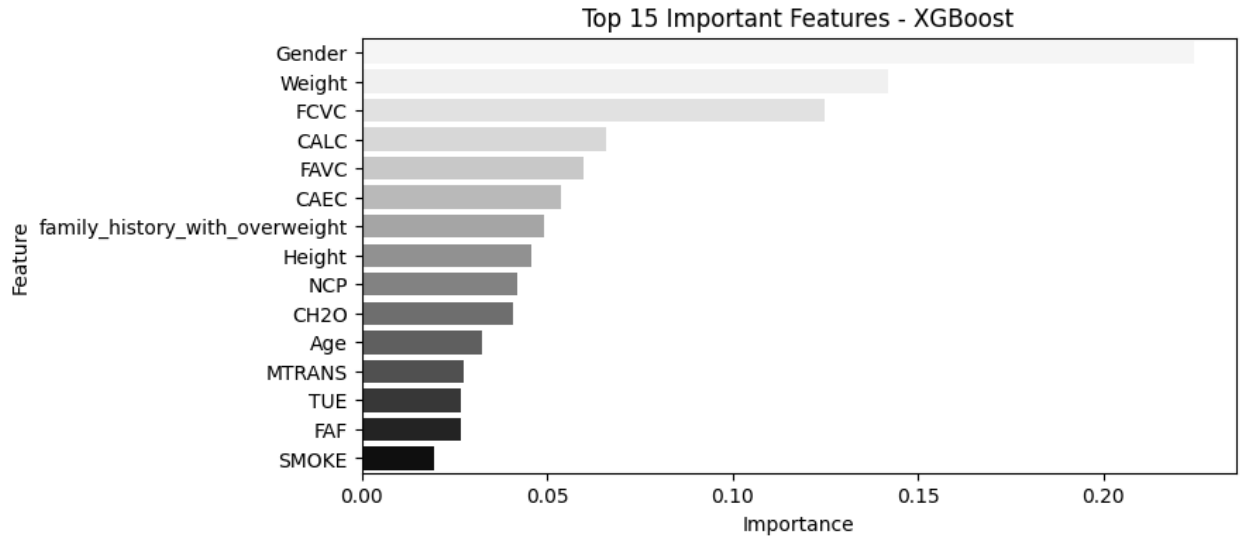
```
Classification Report:
                     precision    recall  f1-score   support

Insufficient_Weight       1.00      0.91      0.95        54
      Normal_Weight       0.85      0.98      0.91        58
     Obesity_Type_I       0.97      0.97      0.97        70
    Obesity_Type_II       0.98      0.98      0.98        60
   Obesity_Type_III       1.00      0.98      0.99        65
 Overweight_Level_I       0.98      0.91      0.95        58
Overweight_Level_II       0.97      0.98      0.97        58

           accuracy                           0.96       423
          macro avg       0.96      0.96      0.96       423
       weighted avg       0.97      0.96      0.96       423
```

In [ ]:

```
cm_xgb = confusion_matrix(y_test, y_pred_xgb)
plt.figure(figsize=(6,5))
sns.heatmap(cm_xgb, annot=True, fmt='d', cmap='Reds',
            xticklabels=target_encoder.classes_,
            yticklabels=target_encoder.classes_)
plt.title("XGBoost Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



In [ ]:
```
importance_xgb = xgb_model.feature_importances_
feat_imp_xgb = pd.DataFrame({'Feature': X_train.columns,'Importance': importar
plt.figure(figsize=(8,4))
sns.barplot(x='Importance', y='Feature', data=feat_imp_xgb.head(15), palette='
plt.title("Top 15 Important Features - XGBoost")
plt.show()
```



Algorithm 4 >> Blended Ensemble (SVM + RF + XGBoost)

In [ ]:
```
svm_probs = svm_model.predict_proba(X_test)
rf_probs = rf_model.predict_proba(X_test)
xgb_probs = xgb_model.predict_proba(X_test)
```

In [ ]:
```
blend_probs = (0.2 * svm_probs) + (0.3 * rf_probs) + (0.5 * xgb_probs)
```

In [ ]:
```
y_pred_blend = np.argmax(blend_probs, axis=1)
```

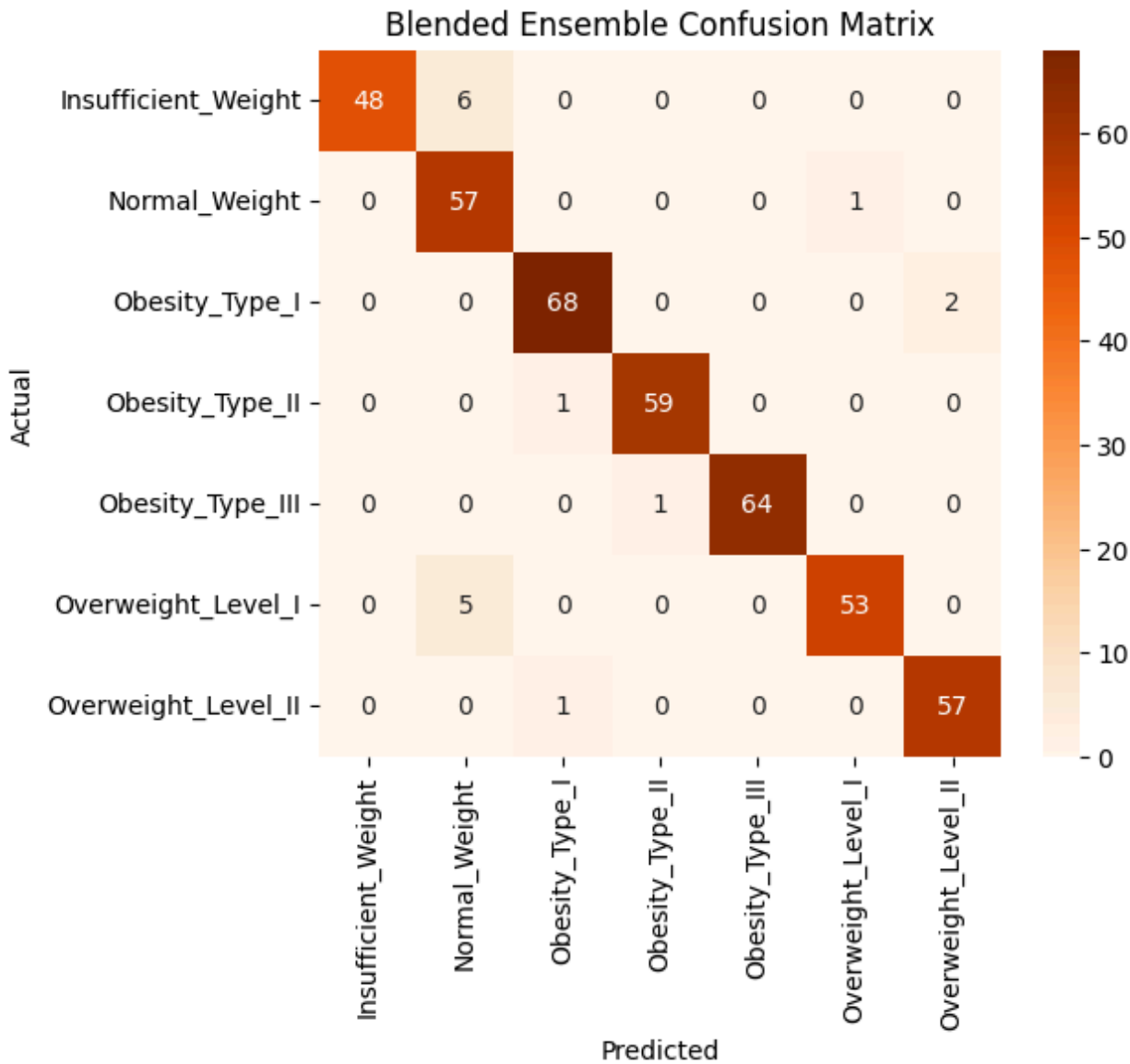
In [ ]:
```
acc_blend = accuracy_score(y_test, y_pred_blend)
print(f"Blended Model Accuracy: {acc_blend:.4f}")
```

```
Blended Model Accuracy: 0.9598
```

In [ ]:
```
print("Classification Report (Blended Model):")
print(classification_report(y_test, y_pred_blend, target_names=target_encoder.
```

```
Classification Report (Blended Model):
                     precision    recall  f1-score   support

Insufficient_Weight       1.00      0.89      0.94        54
      Normal_Weight       0.84      0.98      0.90        58
     Obesity_Type_I       0.97      0.97      0.97        70
    Obesity_Type_II       0.98      0.98      0.98        60
   Obesity_Type_III       1.00      0.98      0.99        65
 Overweight_Level_I       0.98      0.91      0.95        58
Overweight_Level_II       0.97      0.98      0.97        58

           accuracy                           0.96       423
          macro avg       0.96      0.96      0.96       423
       weighted avg       0.96      0.96      0.96       423
```

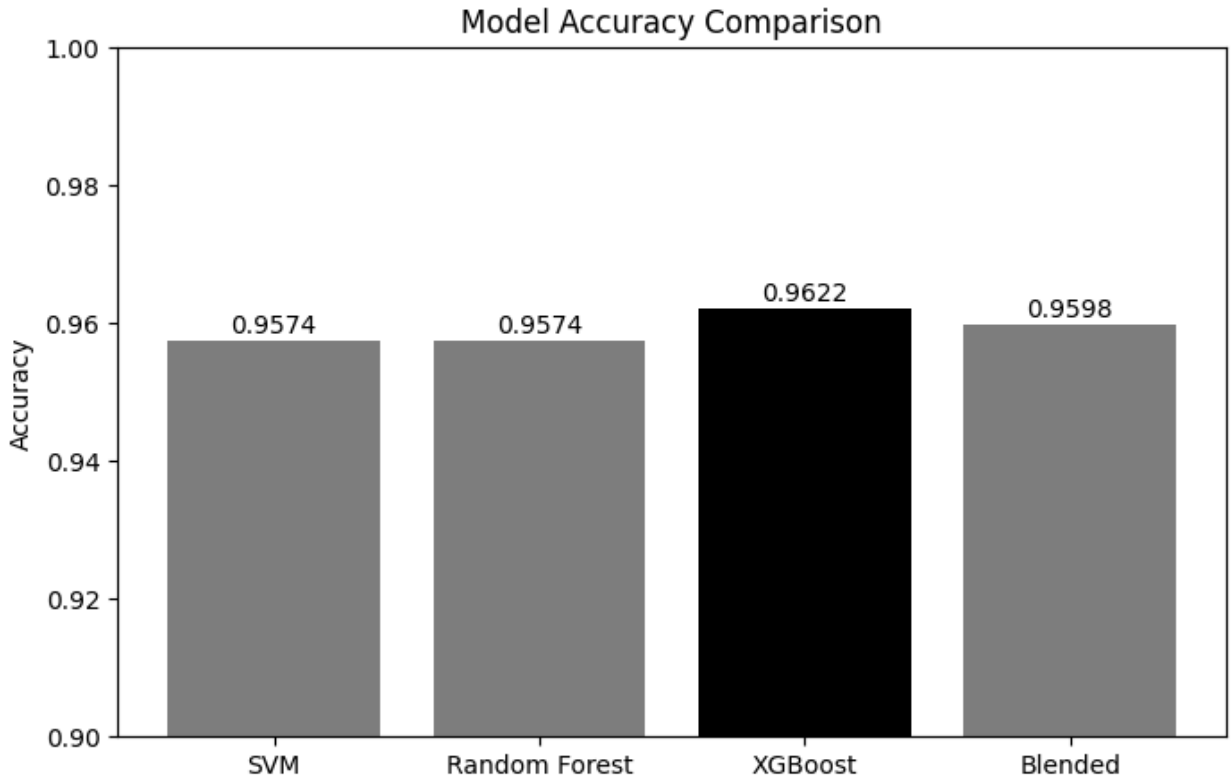
In [ ]:
```
cm_blend = confusion_matrix(y_test, y_pred_blend)
plt.figure(figsize=(6,5))
sns.heatmap(cm_blend, annot=True, fmt='d', cmap='Oranges',
            xticklabels=target_encoder.classes_,
            yticklabels=target_encoder.classes_)
plt.title("Blended Ensemble Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



>> Model Accuracy Comparison

In [ ]:
```
models = ['SVM', 'Random Forest', 'XGBoost', 'Blended']
accuracies = [acc_svm, acc_rf, acc_xgb, acc_blend]
plt.figure(figsize=(8,5))
plt.bar(models, accuracies, color=['grey','grey','black','grey'])
plt.ylim(0.90,1.0)
```

```
plt.title("Model Accuracy Comparison")
plt.ylabel("Accuracy")
for i, v in enumerate(accuracies):
    plt.text(i, v + 0.001, f"{v:.4f}", ha='center')
plt.show()
```



OBJECTIVE 2 > FEATURE SELECTION VS MODEL COMPLEXITY TRADE-OFFS

>> Most Important Feature Selection Table - XGBoost

In [ ]:
```
feat_imp_xgb = pd.DataFrame({'Feature': X_train.columns,'Importance': xgb_mode
print("Feature Ranking Based on XGBoost Importance:\n")
print(feat_imp_xgb)
```

```
Feature Ranking Based on XGBoost Importance:

                            Feature  Importance
1                            Gender    0.224452
3                            Weight    0.141923
6                              FCVC    0.125022
4                              CALC    0.066091
5                              FAVC    0.059682
14                             CAEC    0.053615
11  family_history_with_overweight    0.049072
2                            Height    0.045932
7                               NCP    0.041834
10                            CH2O    0.040750
0                              Age    0.032452
15                           MTRANS    0.027403
13                              TUE    0.026806
12                              FAF    0.026760
9                             SMOKE    0.019668
8                               SCC    0.018539
```

>> Model Complexity Trade-Off : Effect of Feature Reduction on Accuracy

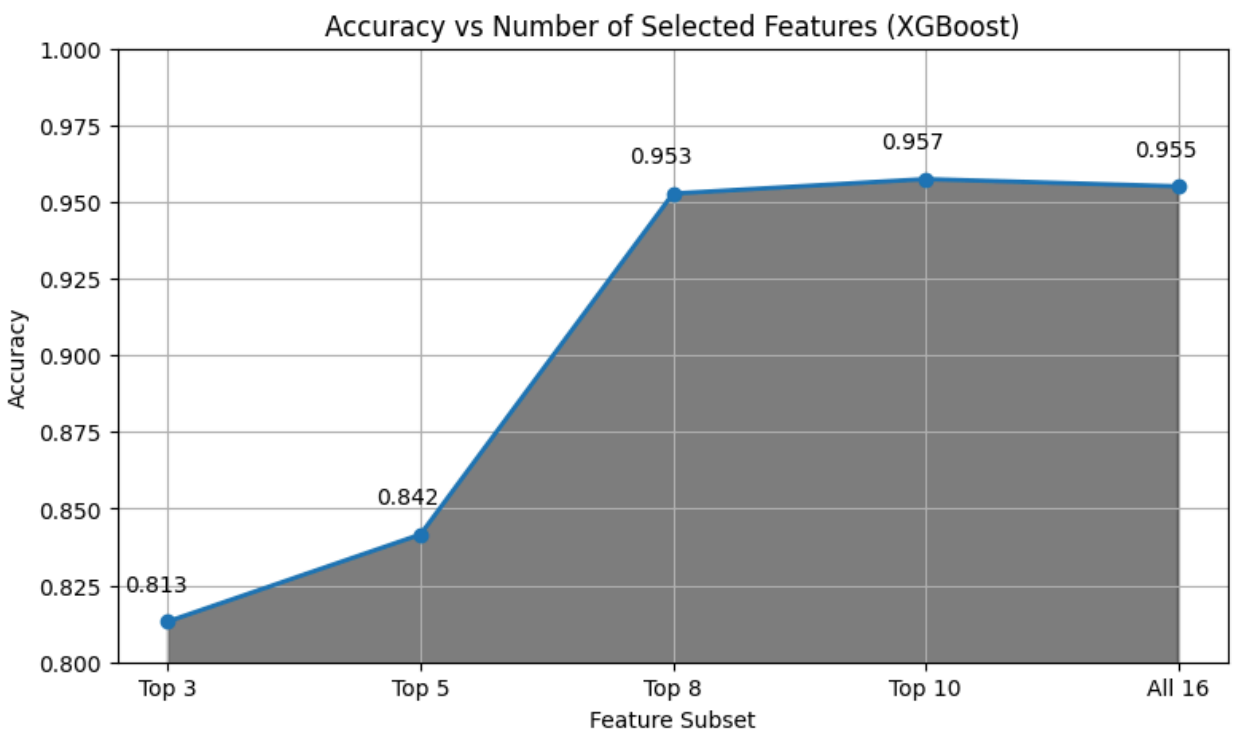
In [ ]:

```python
feature_ranks = feat_imp_xgb['Feature'].tolist()
subset_sizes = [3, 5, 8, 10, len(feature_ranks)]
subset_names = ["Top 3", "Top 5", "Top 8", "Top 10", "All 16"]
subset_accuracies = []
print("Testing XGBoost With Different Feature Subsets:\n")
for size in subset_sizes:
    selected_features = feature_ranks[:size]

    X_train_sub = X_train[selected_features]
    X_test_sub = X_test[selected_features]

    model = XGBClassifier(
        n_estimators=300,
        learning_rate=0.05,
        max_depth=6,
        subsample=0.9,
        colsample_bytree=0.9,
        objective='multi:softmax',
        num_class=len(target_encoder.classes_),
        eval_metric='mlogloss',
        random_state=42
    )
    model.fit(X_train_sub, y_train)
    preds = model.predict(X_test_sub)
    acc = accuracy_score(y_test, preds)
    subset_accuracies.append(acc)
    print(f"{size} features ({subset_names[subset_sizes.index(size)]}): Accura
```

```
Testing XGBoost With Different Feature Subsets:

3 features (Top 3): Accuracy = 0.8132
5 features (Top 5): Accuracy = 0.8416
8 features (Top 8): Accuracy = 0.9527
10 features (Top 10): Accuracy = 0.9574
16 features (All 16): Accuracy = 0.9551
```

In [ ]:

```
plt.figure(figsize=(9,5))
plt.plot(subset_names, subset_accuracies, marker='o', linestyle='-', linewidth
plt.fill_between(subset_names, subset_accuracies, color='black', alpha=0.5)
plt.ylim(0.8,1)
plt.title("Accuracy vs Number of Selected Features (XGBoost)")
plt.xlabel("Feature Subset")
plt.ylabel("Accuracy")
for i, v in enumerate(subset_accuracies):
    plt.text(i-0.05, v+0.01, f"{v:.3f}", ha='center')
plt.grid(True)
plt.show()
```



>> Performance Comparison Table - All Models & Feature Subsets

Models	Features Used	Accuracy	Notes
> SVM (RBF)	All 16	0.9574	Strong baseline, performs well with scaling

Models	Features Used	Accuracy	Notes
> Random Forest	All 16	0.9574	Good interpretability via feature importance
>> XGBoost	All 16	**0.9622**	Best overall performer
> Blended Ensemble (SVM+RF+XGB)	All 16	0.9598	Slightly below XGBoost - blending didn’t improve

Top Feature Subsets - XGBoost	Features Used	Accuracy	Notes
> Top 3 Features - XGBoost	Top 3	0.8132	Insufficient features - poor accuracy
> Top 5 Features - XGBoost	Top 5	0.8416	Still low - model too simplistic
> Top 8 Features - XGBoost	Top 8	0.9527	Very close to full accuracy - ideal trade-off
> Top 10 Features - XGBoost	Top 10	0.9574	Nearly best - diminishing returns begin
> Top 16 Features - XGBoost	All features	0.9551	Slightly below best due to weaker features noise